

K-zero:

boundary propagators

$$X_{k+1,j} = X_{1,j} - X_{1,k+2} \quad j = k+3, \dots, n-1$$

$$X_{k+1,n} = -X_{1,k+2}$$

$$X_{l,j} = X_{k+1,j} + X_{l,k+2} \quad l = 2, \dots, k \\ j = k+3, \dots, n$$

$$\# \text{ variables: } (k+1)(n-k-2) + (k-1)$$

$$\# \text{ equations: } k(n-k-2)$$

$$\# \text{ independents: } \boxed{n-3}$$

if $1, k+2, k+3, \dots, n-1$ are independent: $n-k-2$ things

for each $l=2, \dots, k$ we can get $X_{l,k+2}$ as independent: $k-1$ things

since we must always send $X_{1,k+2}$ and $X_{k+1,n}$ to ∞ , we can have at most $n-4$ chosen-free.

ex: $n=8$
 $k=3$

$$\boxed{X_{46}} = X_{16} - X_{15}$$

$$\boxed{X_{47}} = X_{17} - X_{15}$$

$$\boxed{X_{48}} = -X_{15}$$

$$\boxed{X_{26}} = X_{16} - X_{15} + X_{25}$$

$$\boxed{X_{27}} = X_{17} - X_{15} + X_{25}$$

$$\boxed{X_{28}} = -X_{15} + X_{25}$$

$$\boxed{X_{36}} = X_{16} - X_{15} + X_{35}$$

$$\boxed{X_{37}} = X_{17} - X_{15} + X_{35}$$

$$\boxed{X_{38}} = -X_{15} + X_{35}$$

fix $(15, 16, 17, 25, 35)$

$\Rightarrow$ fix everything else

chosen free for $k=3$

$$\left. \begin{aligned} X_{4j} &= X_{1j} - X_{15} \\ X_{3j} &= X_{4j} + X_{35} \\ X_{2j} &= X_{4j} + X_{25} \end{aligned} \right\} j = 6, \dots, n-1$$

$$\begin{aligned} X_{4n} &= -X_{15} \\ X_{3n} &= -X_{15} + X_{35} \\ X_{2n} &= -X_{15} + X_{25} \end{aligned}$$

$$\begin{aligned} X_{3j} &= X_{1j} + X_{3n} \\ X_{2j} &= X_{1j} + X_{2n} \end{aligned}$$

- if X_{25} is CF $\leftarrow$ chosen free then it must be $4j, 2j \rightarrow \infty \quad \forall j \in \{6, \dots, n-1\}$
 - therefore we can choose $1j$ to be also independent.
- if X_{2n} is CF then it must be $4j, 2j \rightarrow \infty \quad \forall j$
 - therefore we can choose $4j$ to be also independent.
- if X_{35} is CF then it must be $4j, 3j \rightarrow \infty \quad \forall j \in \{6, \dots, n-1\}$
 - therefore we can choose $1j$ to be also independent.
- if X_{3n} is CF then it must be $4j, 3j \rightarrow \infty \quad \forall j$
 - therefore we can choose $4j$ to be also independent.

options

① $25, 35, 1j \quad V_j \quad CF$

② $2n, 3n, 4j \quad V_j \quad CF$

③ ^{new} $25, 3n \quad CF$

④ $2n, 35 \quad CF$

⑤ 25 and one from each $(1j, 3j)$

⑥ $2n$ and one from each $(4j, 3j)$

⑦ 35 and one from each $(1j, 2j)$

⑧ $3n$ and one from each $(4j, 2j)$

→ send both X_{1j} and X_{4j} to ∞ !

$$\left. \begin{aligned} X_{4j} &= X_{1j} - X_{15} \\ X_{3j} &= X_{4j} + X_{35} \\ X_{2j} &= X_{4j} + X_{25} \end{aligned} \right\} j=6, \dots, n-1$$

$$X_{4n} = -X_{15}$$

$$X_{3n} = -X_{15} + X_{35}$$

$$X_{2n} = -X_{15} + X_{25}$$

$$\begin{aligned} X_{3j} &= X_{1j} + X_{3n} \\ X_{2j} &= X_{1j} + X_{2n} \end{aligned}$$

$$X_{15} = \lambda \rightarrow \infty$$

$$X_{1j} = 2\lambda \rightarrow \infty$$

$$X_{4j} = \lambda \rightarrow \infty$$

$$X_{2j} = X_{4j} + X_{25} \rightarrow \infty$$

$$X_{3j} = X_{1j} + X_{3n} \rightarrow \infty$$

All that remains are the cases in which $25, 2n, 35, 3n$ all go to ∞ .

for each j :

$$X_{15} = \lambda_1$$

$$X_{25} = \lambda_2$$

$$X_{35} = \lambda_3$$

$$X_{4n} = -\lambda_1$$

$$X_{2n} = \lambda_2 - \lambda_1$$

$$X_{3n} = \lambda_3 - \lambda_1$$

• $X_{4j} = X_{1j} - \lambda_1$: at most one of $1j, 4j$ is CF

• $X_{2j} = X_{4j} + X_{25}$: at most one of $2j, 4j$ is CF

• $X_{2j} = X_{1j} + X_{2n}$: at most one of $2j, 1j$ is CF

• $X_{3j} = X_{4j} + X_{35}$: at most one of $3j, 4j$ is CF

• $X_{3j} = X_{1j} + X_{3n}$: at most one of $3j, 1j$ is CF

if $1j$ CF, $2j, 3j, 4j \rightarrow \infty$

if $4j$ CF, $2j, 3j, 1j \rightarrow \infty$

what if we choose both $2j, 3j$ and send $1j, 4j \rightarrow \infty$

$$X_{15} = \lambda$$

$$X_{2j} = \alpha$$

$$X_{3j} = \beta$$

$$X_{4j} = \alpha - X_{25} = \beta - X_{35}$$

$$X_{1j} = \alpha - X_{2n} = \beta - X_{3n}$$

so $\begin{cases} X_{25} = \alpha - \beta + X_{35} \\ X_{2n} = \alpha - \beta + X_{3n} \end{cases}$

ex: $X_{35} = 2\lambda$

$$X_{3n} = \lambda$$

$$X_{25} = \alpha - \beta + 2\lambda$$

$$X_{2n} = \alpha - \beta + \lambda$$

$$\left. \begin{aligned} X_{4j} &= X_{1j} - X_{15} \\ X_{3j} &= X_{4j} + X_{35} \\ X_{2j} &= X_{4j} + X_{25} \end{aligned} \right\} j=6, \dots, n-1$$

$$X_{4n} = -X_{15}$$

$$X_{3n} = -X_{15} + X_{35}$$

$$X_{2n} = -X_{15} + X_{25}$$

$$\begin{aligned} X_{3j} \\ X_{2j} \end{aligned}$$

Can we do $X_{2j_1}, X_{3j_1}, X_{2j_2}, X_{3j_2}$ CF?

$$X_{15} = \lambda$$

$$X_{4j_1} = \alpha_1 - X_{25} = \beta_1 - X_{35}$$

$$X_{4j_2} = \alpha_2 - X_{25} = \beta_2 - X_{35}$$

$$X_{2j_1} = \alpha_1$$

$$X_{3j_1} = \beta_1$$

$$X_{2j_2} = \alpha_2$$

$$X_{3j_2} = \beta_2$$

$$X_{25} = \alpha_1 - \beta_1 + X_{35}$$

$$X_{25} = \alpha_2 - \beta_2 + X_{35}$$

$$\Rightarrow \alpha_1 - \beta_1 = \alpha_2 - \beta_2 \rightarrow \text{cannot be all independent!}$$

Figure out the possible sets of independents first!

(should be a linear algebraic thing)

Then we can look at what can have to go to ∞ .

Note: if we pick $2j, 3j \forall j \in \{6, \dots, n-1\}$ as indep. we get $2 \cdot (n-6)$ things but we can have $n-4$ at most, so it makes perfect sense.

expectation: pick both $2j$ and $3j$ for a specific j , then choose $1j$ or $4j$ for the others, which gives $n-5$ CF.